Data Structures

LECTURE-2

What is an Array?

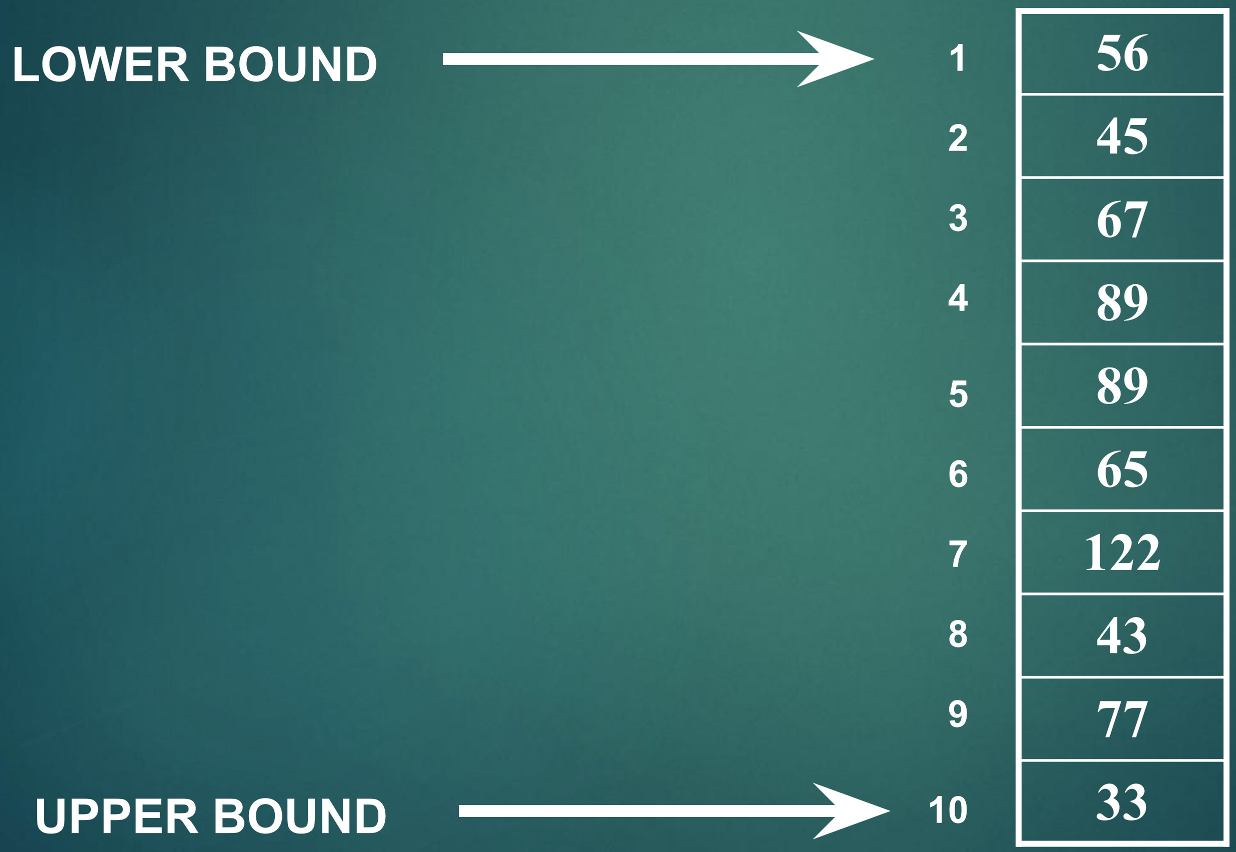
A block of memory divided into separate storage locations. Each storage location is capable of holding a single value of a specific type. All values in the array must be of the same type.

The elements of the array are referenced respectively by an index set consisting of n consecutive numbers.

TYPES OF ARRAYS

- SINGLE DIMENSION ARRAY
- DOUBLE DIMENSION ARRAY
- MULTI-DIMENSION ARRAY

- UB is the largest index called the upper bound
- LB is the smallest index called the lower bound



LINEAR ARRAY

ALGORITHMS

Algorithm-1

Traversing a Linear Array using for loop

1. Repeat for K = LB to UB
2. [visit element] Apply process to LA(K)

   [End of loop]
3. Exit

Algorithm-2

Traversing a Linear Array using while loop

1. [initialize counter] set k:=LB
2. Repeat steps 3 and 4 while K ≤ UB
3. [visit element] Apply process to LA[K]
4. [increase counter] set k:=k+1

   [end of loop]
5. Exit

Algorithm-3

Inserting into a Linear Array

INSERT(LA, N, K, ITEM)

1. [initialize counter] set J:=N
2. Repeat steps 3 and 4 while J ≥ K
3. [move jth element downward] set LA[J+1]:=LA[J]
4. [decrease counter] set J:=J-1

   [end of loop]
5. [insert element] set LA[K]:=ITEM
6. [reset N] set N:=N+1
7. Exit

Algorithm-4

Deleting from a Linear Array

DELETE(LA, N, K, ITEM)

1. set ITEM=LA[K]
2. Repeat for J = K to N-1

   set LA[J]:=LA[J+1]

   [end of loop]
3. set N:=N-1
4. Exit

What is Bubble Sort?

A multiple-pass sorting technique that starts by sequencing the first two items, then the second with the third, then the third with the fourth, and so on, until the end of the set has been reached. The process is repeated until all items are in the correct sequence.

Algorithm-5

Bubble Sort

BUBBLE(DATA, N)

1. Repeat steps 2 and 3 for K = 1 to N-1
2. Set PTR :=1
3. Repeat while PTR ≤ N-K
   a) if DATA[PTR] > DATA[PTR+1], then
      interchange DATA[PTR] and DATA[PTR+1]
      [end of if structure]
   b) set PTR :=PTR+1
   [end of innter loop]
[end of outer loop]
4. Exit

PRACTICE

Example 4.7

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What is Linear Search?

Also known as a sequential search, linear search is a method of how a search is performed. With a linear search each item is examined, one at a time, in sequence, until a matching result is found.

Algorithm-6

Linear Search

LINEAR(DATA, N, ITEM, LOC)

1. [insert item at the end of DATA] set DATA[N+1] :=ITEM
2. Set LOC :=1
3. Repeat while DATA[LOC] ≠ ITEM

   set LOC :=LOC+1

   [end of loop]
4. Display "Value found at Location" LOC
5. Exit

What is Binary Search?

Binary search is a searching technique for searching a set of sorted data for a particular value.

- Start search in the middle of array
- if x is less than the middle element, search in lower half
- if x is greater than the middle element, search in upper half

Algorithm-7

Binary Search

BINARY(DATA, LB, UB, ITEM, LOC)

1. [initialize variables]
   BEG :=LB, END :=UB and MID=INT((BEG+END)/2)
2. Repeat steps 3 & 4 while BEG ≤ END and DATA[MID] ≠ ITEM
3. if ITEM < DATA[MID], then:
   set END :=MID-1
   else:
   set BEG :=MID+1
   [end of if structure]
4. Set MID=INT((BEG+END)/2)
   [end of loop]
5. If DATA[MID] = ITEM, then:
   set LOC :=MID
   else:
   set LOC :=NULL
   [end of if structure]
6. Exit.

PRACTICE

Example 4.9

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Double Dimension Array

A two dimensional m X n array is a collection of data elements such that each element is specified by a pair of integers called subscripts

Two dimension array are called matrices in mathematics and tables in business applications.

Two dimension array are sometimes called matrix arrays.

Double Dimension Array…(cont)

There is a standard way of drawing a two-dimensional m x n array where the elements form a rectangular array with m rows and n columns.

3 x 4 Array

	1	2	3	4
1				
2				
3				

Double Dimension Array…(cont)

Storing the array column by column is called **column-major order.**

Storing the array row by row is called **row-major order.**